

Homework and Tutorial

Assignments

➤ **4.14, 4.25, 4.31, 4.33, 4.35**

Tutorial problems

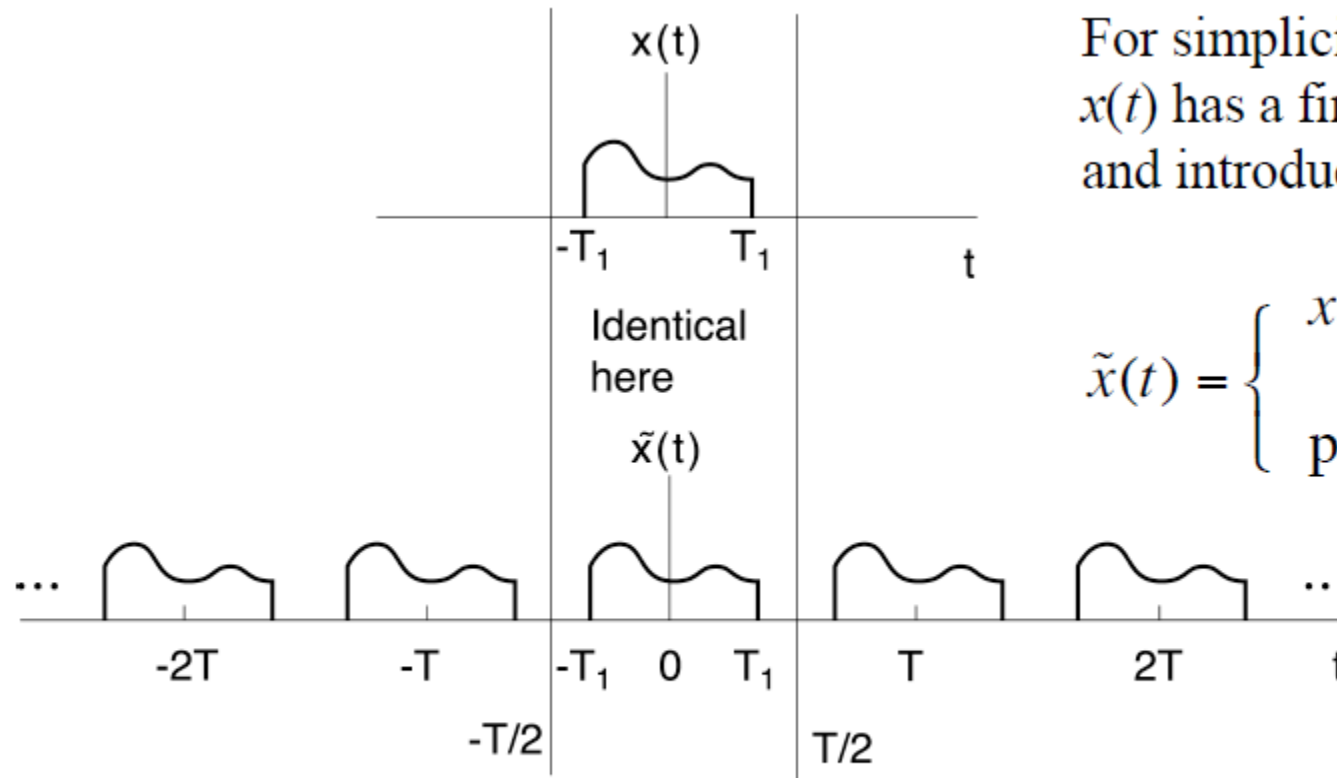
- **Basic Problems with Answers 4.10, 4.16**
- **Basic Problems 4.26, 4.30**

Chapter 4

The Continuous-Time Fourier Transform

(cont.)

So, on the derivation of FT ...



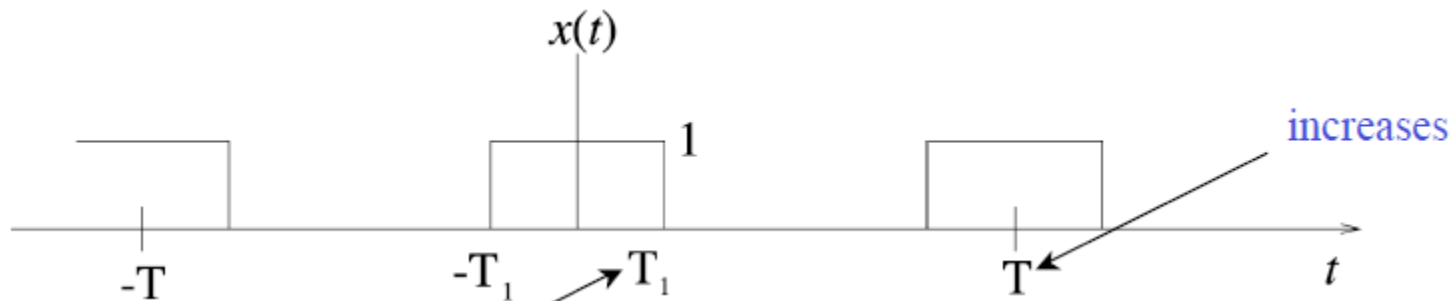
For simplicity, assume $x(t)$ has a finite duration, and introduce a periodic $\tilde{x}(t)$

$$\tilde{x}(t) = \begin{cases} x(t) & -\frac{T}{2} < t < \frac{T}{2} \\ \text{periodic} & |t| > \frac{T}{2} \end{cases}$$

As $T \rightarrow \infty$, $x(t) = \tilde{x}(t)$ for all t

Review

Motivating Examples: Square wave



$$a_k = \frac{2 \sin(k\omega_0 T_1)}{k\omega_0 T}$$

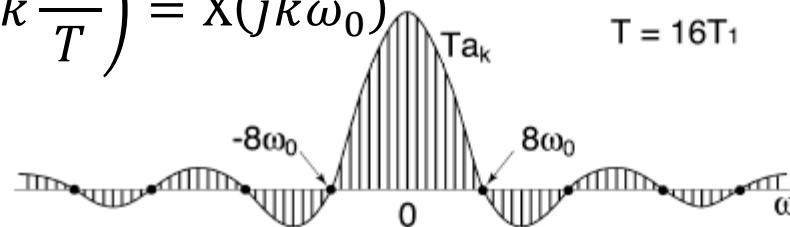
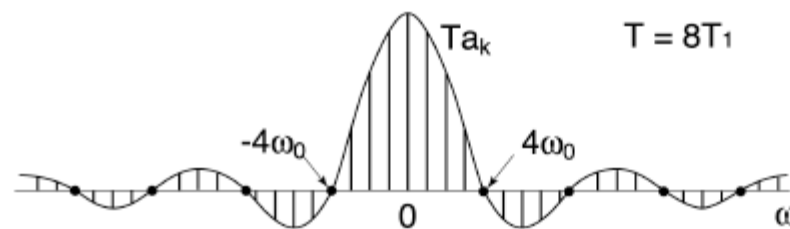
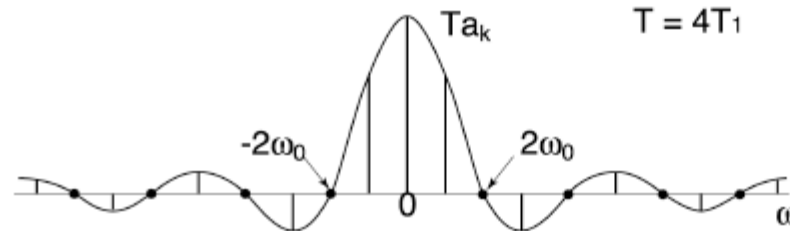
keep fixed

↓

$$T a_k = \frac{2 \sin(k\omega_0 T_1)}{k\omega_0}$$

(independent of T)

$$\text{Let } X(j\omega) = \frac{2 \sin \omega T_1}{\omega}, T a_k = X\left(jk \frac{2\pi}{T}\right) = X(jk\omega_0)$$



$$\left(\omega_0 = \frac{2\pi}{T}\right)$$

Become denser in ω as T increases



Derivation (cont.): Analysis equation

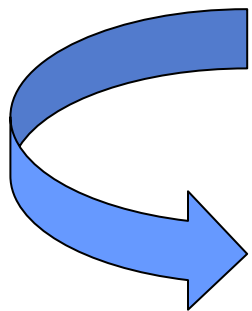
$$\tilde{x}(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} \quad \omega_o = \frac{2\pi}{T}$$

$$a_k = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} \tilde{x}(t) e^{-jk\omega_0 t} dt = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t) e^{-jk\omega_0 t} dt$$



$\tilde{x}(t) = x(t)$ in this interval

$$= \frac{1}{T} \int_{-\infty}^{+\infty} x(t) e^{-jk\omega_0 t} dt \quad (1)$$



If we define

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$$

then, Eq. (1) $\Rightarrow$

$$a_k = \frac{1}{T} X(jk\omega_0) = \frac{1}{T} X(j\omega) |_{\omega=k\omega_0}$$

The CT Fourier Transform Pair

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t)e^{-j\omega t} dt \quad \text{— } FT$$

Fourier Transform

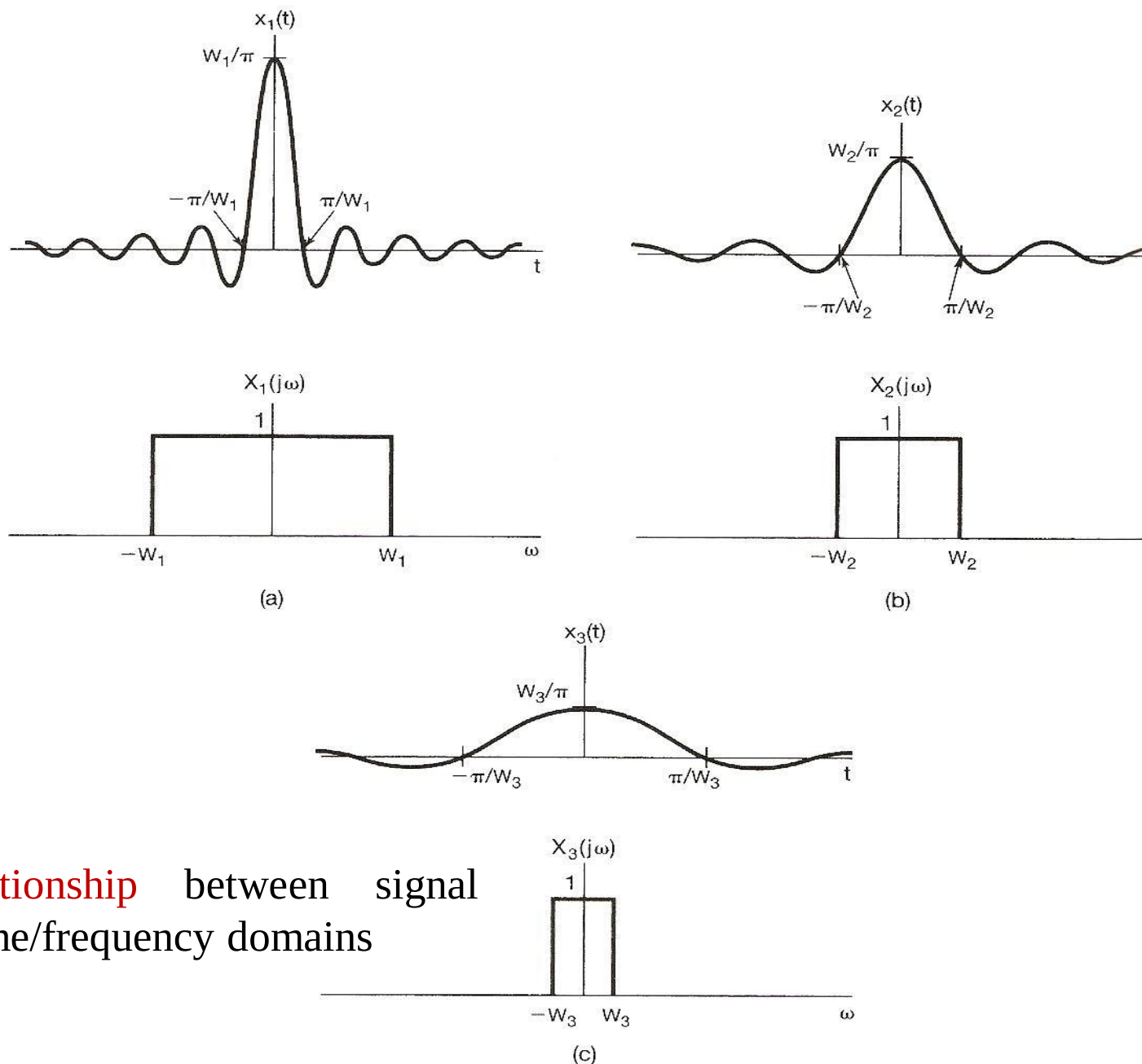
$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega)e^{j\omega t} d\omega \quad \text{— Inverse } FT$$

Inverse Fourier Transform

$$x(t) \xleftrightarrow{\mathcal{F}} X(j\omega)$$

$$\mathcal{F}(x(t)) = X(j\omega)$$

$$x(t) = \mathcal{F}^{-1}(X(j\omega))$$



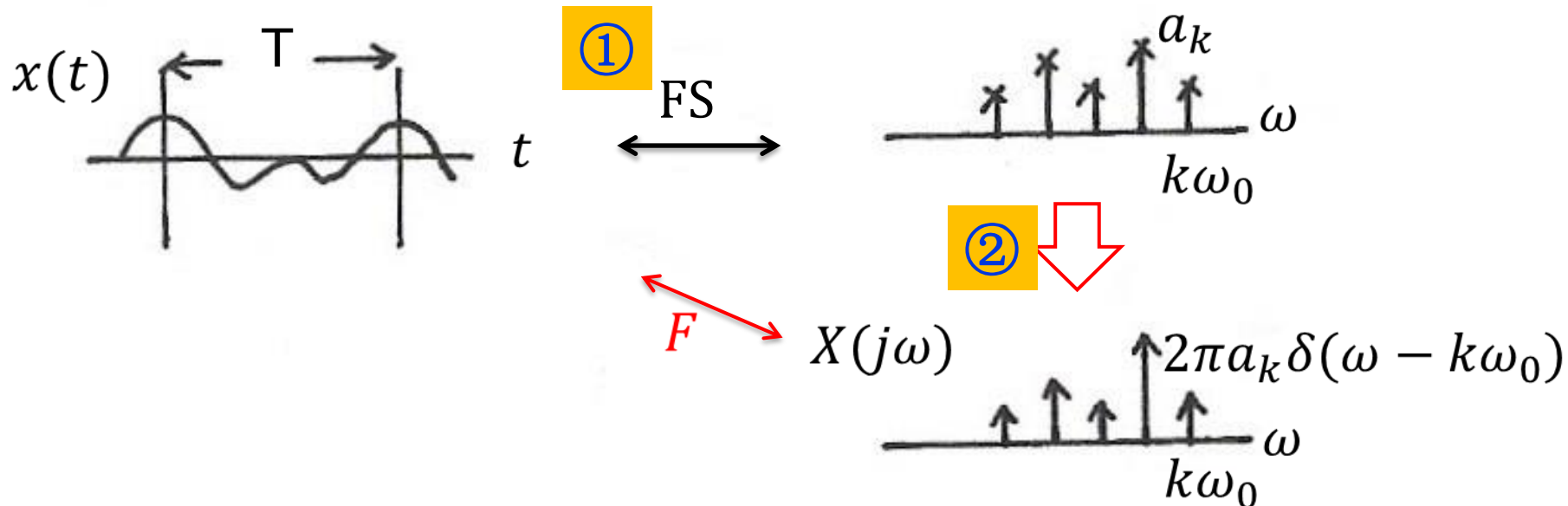
Inverse relationship between signal
“width” in time/frequency domains

Figure 4.11 Fourier transform pair of Figure 4.9 for several different values of W .

Fourier Transform for Periodic Signals – Unified Framework

More generally, if $x(t) = x(t+T)$, then

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} \longleftrightarrow X(j\omega) = \sum_{k=-\infty}^{+\infty} 2\pi a_k \delta(\omega - k\omega_0) \quad \text{Discrete spectra}$$



Outline

- Interesting properties of CTFT
 - Real signal
 - Convolution property – LTI systems
 - Multiplication property – Modulation & Sampling
- Linear-constant-coefficient differential equation of LTI systems

$$x(t) \longleftrightarrow X(j\omega)$$

CTFT Properties (cont.)

- Time reversal

$$x(-t) \longleftrightarrow X(-j\omega)$$

- Conjugate Symmetry

$$x(t) \text{ real} \longleftrightarrow X(-j\omega) = X^*(j\omega)$$

$$|X(-j\omega)| = |X(j\omega)|$$

Even

Or

$$\text{Re}\{X(-j\omega)\} = \text{Re}\{X(j\omega)\}$$

Even

$$\angle X(-j\omega) = -\angle X(j\omega)$$

Odd

$$\text{Im}\{X(-j\omega)\} = -\text{Im}\{X(j\omega)\}$$

Odd

$\Downarrow$

a) $x(t)$ real and even

$$x(t) = x(-t) = x^*(t)$$

$$\Rightarrow X(j\omega) = X(-j\omega) = X^*(j\omega) \text{ — Real \& even}$$

b) $x(t)$ real and odd

$$x(t) = -x(-t) = x^*(t)$$

$$\Rightarrow X(j\omega) = -X(-j\omega) = -X^*(j\omega) \text{ — Purely imaginary \& odd}$$

$$\text{c) } X(j\omega) = \text{Re}\{X(j\omega)\} + j\text{Im}\{X(j\omega)\}$$

$$\begin{array}{c} \uparrow \quad \quad \uparrow \quad \quad \uparrow \\ \text{For real } x(t) = \text{Ev}\{x(t)\} + \text{Od}\{x(t)\} \end{array}$$

Properties of the CT Fourier Transform

1) Linearity

$$x(t) \xleftrightarrow{F} X(j\omega), y(t) \xleftrightarrow{F} Y(j\omega)$$

$$ax(t) + by(t) \xleftrightarrow{F} aX(j\omega) + bY(j\omega)$$

2) Time Shifting

$$x(t - t_0) \longleftrightarrow e^{-j\omega t_0} X(j\omega)$$

Proof:
$$\int_{-\infty}^{\infty} x(\underbrace{t - t_0}_{t'}) e^{-j\omega t} dt = e^{-j\omega t_0} \underbrace{\int_{-\infty}^{\infty} x(t') e^{-j\omega t'} dt'}_{X(j\omega)}$$

FT magnitude unchanged

$$|e^{-j\omega_0 t} X(j\omega)| = |X(j\omega)|$$

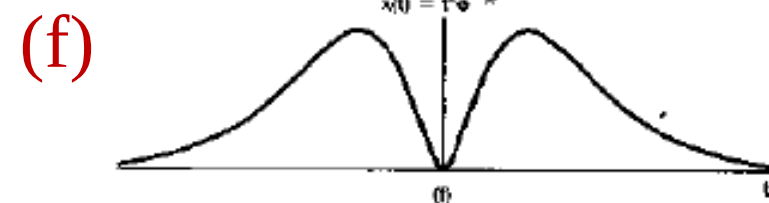
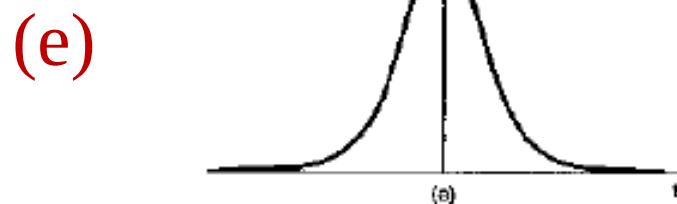
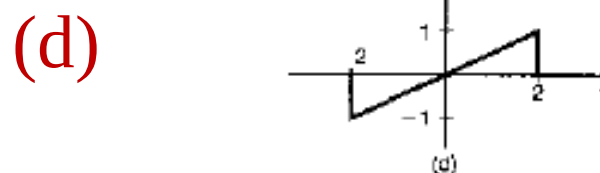
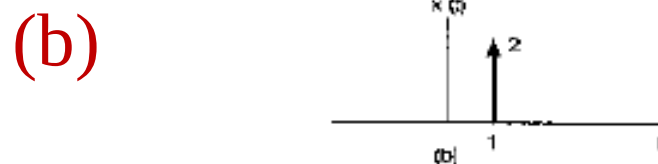
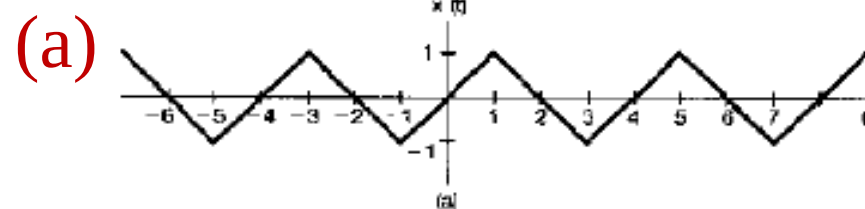
Linear change in FT phase

$$\angle(e^{-j\omega_0 t} X(j\omega)) = \angle X(j\omega) - \omega t_0$$

Problem 4.24 (a)

- Determine which, if any, of the real signals in (a)-(f) have Fourier transforms that satisfy each of the following condition:

- $\text{Re}\{X(j\omega)\} = 0$
- $\text{Im}\{X(j\omega)\} = 0$
- There exists a real a such that $e^{ja\omega}X(j\omega)$ is real
- $\int_{-\infty}^{\infty} X(j\omega)d\omega = 0$
- $\int_{-\infty}^{\infty} \omega X(j\omega)d\omega = 0$
- $X(j\omega)$ is periodic



Impulse & Frequency Responses

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \longrightarrow \boxed{h(t)} \longrightarrow y(t) = \sum_{k=-\infty}^{\infty} a_k H(jk\omega_0) e^{jk\omega_0 t}$$

$$H(j\omega) = \int_{-\infty}^{+\infty} h(t) e^{-j\omega t} dt$$

Impulse response $\xleftrightarrow{\mathcal{F}}$ Frequency response

Remark:

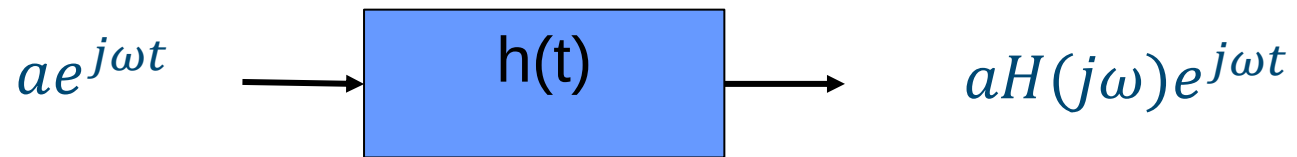
Not every LTI system has frequency response; but according to Dirichlet conditions, most of stable LTI systems have.

Convolution Property

$$y(t) = h(t) * x(t) \leftrightarrow Y(j\omega) = H(j\omega)X(j\omega)$$

An intuitive interpretation:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega \Rightarrow x(t) = \int_{-\infty}^{+\infty} \underbrace{\frac{1}{2\pi} X(j\omega) d\omega}_{\text{coefficient}} e^{j\omega t}$$



$$y(t) = \int_{-\infty}^{+\infty} \underbrace{H(j\omega) \frac{1}{2\pi} X(j\omega) d\omega}_{\text{new coefficient}} e^{j\omega t} \Rightarrow y(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} H(j\omega) X(j\omega) e^{j\omega t} d\omega$$

$$Y(j\omega) = H(j\omega) X(j\omega)$$

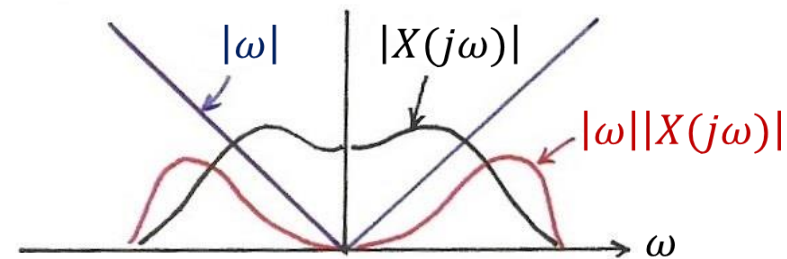
Frequency Response Examples

Example 4.16 A differentiator

$$y(t) = \frac{dx(t)}{dt} \quad \text{— an LTI system}$$

From differentiation property $\Rightarrow \frac{d}{dt} \xleftrightarrow{FT} j\omega$

$$\Downarrow \\ H(j\omega) = j\omega$$



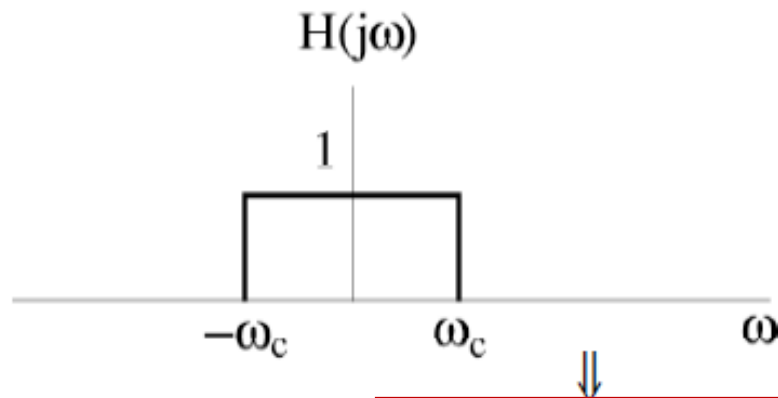
1) Amplifies high frequencies (enhances sharp edges)

2) $+\pi/2$ phase shift ($j = e^{j\pi/2}$) Larger at high ω_0 phase shift

$$\frac{d}{dt} \sin \omega_0 t = \omega_0 \cos \omega_0 t = \omega_0 \sin(\omega_0 t + \pi/2)$$

$$\frac{d}{dt} \cos \omega_0 t = -\omega_0 \sin \omega_0 t = \omega_0 \cos(\omega_0 t + \pi/2)$$

Example 4.18 : Impulse Response of an *Ideal* Lowpass Filter

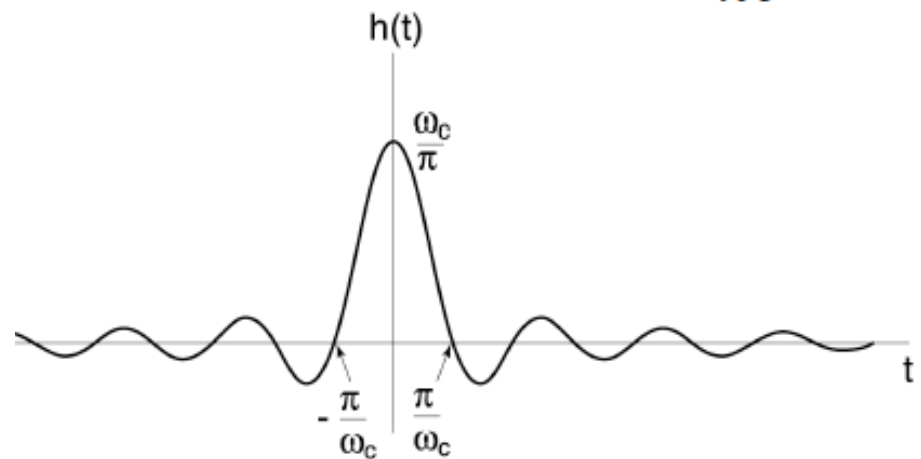


$$h(t) = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega t} d\omega = \frac{\sin \omega_c t}{\pi t} = \frac{\omega_c}{\pi} \operatorname{sinc}\left(\frac{\omega_c t}{\pi}\right)$$

Questions:

- 1) Is this a causal system?
- 2) What is $h(0)$?

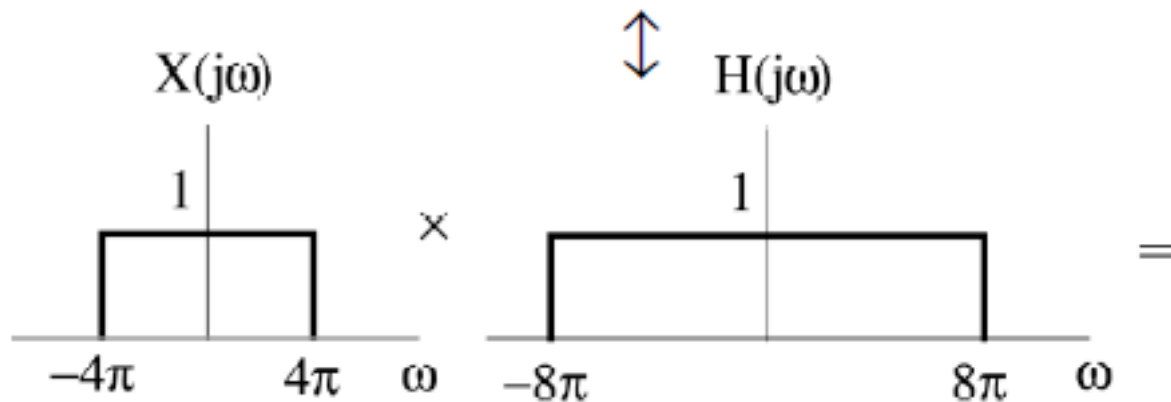
$$\operatorname{sinc}(\theta) = \frac{\sin(\pi\theta)}{\pi\theta}$$



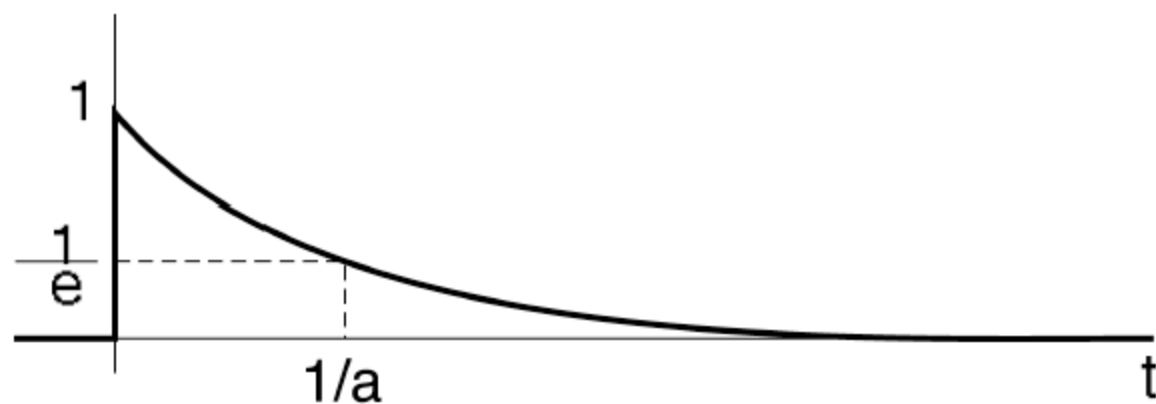
Convolution Calculation Examples

Example 4.20

$$\underbrace{\frac{\sin 4\pi t}{\pi t}}_{x(t)} * \underbrace{\frac{\sin 8\pi t}{\pi t}}_{h(t)} = ?$$



Review from the last lecture, right-sided exponential



$$x(t) = e^{-at}u(t) , \quad a > 0$$

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t)e^{-j\omega t} dt = \int_0^{+\infty} \underbrace{e^{-at}e^{-j\omega t}}_{e^{-(a+j\omega)t}} dt$$

$$= -\left(\frac{1}{a+j\omega}\right)e^{-(a+j\omega)t}\bigg|_0^{\infty} = \boxed{\frac{1}{a+j\omega}}$$

Example 4.19

$$h(t) = e^{-t}u(t) \quad , \quad x(t) = e^{-2t}u(t)$$

$$y(t) = h(t) * x(t) = ?$$

⇓

$$Y(j\omega) = H(j\omega)X(j\omega) = \frac{1}{(1+j\omega)} \cdot \frac{1}{(2+j\omega)}$$

⇓ Partial fraction expansion

$$Y(j\omega) = \frac{1}{1+j\omega} \overset{\mathbf{a=1}}{-} \frac{1}{2+j\omega} \overset{\mathbf{a=2}}{+}$$

⇓ inverse FT

$$y(t) = [e^{-t} - e^{-2t}]u(t)$$

Multiplication Property

Since *FT* is highly symmetric,

$$x(t) = \mathcal{F}^{-1}\{X(j\omega)\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega, \quad X(j\omega) = \mathcal{F}\{x(t)\} = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

thus if

$$x(t) * y(t) \longleftrightarrow X(j\omega) \cdot Y(j\omega)$$

then the other way
around is also true

$$\begin{aligned} x(t) \cdot y(t) &\longleftrightarrow \frac{1}{2\pi} X(j\omega) * Y(j\omega) \\ &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\theta) Y(j(\omega - \theta)) d\theta \end{aligned}$$

$\frac{1}{2\pi}$

— A consequence of *Duality*

Example: Frequency Shifting

$$e^{j\omega_0 t} x(t) \xleftrightarrow{F} X(j(\omega - \omega_0))$$

$$e^{j\omega_0 t} \cdot x(t) \leftrightarrow \frac{1}{2\pi} [2\pi\delta(\omega - \omega_0) * X(j\omega)] \\ = X(j(\omega - \omega_0))$$

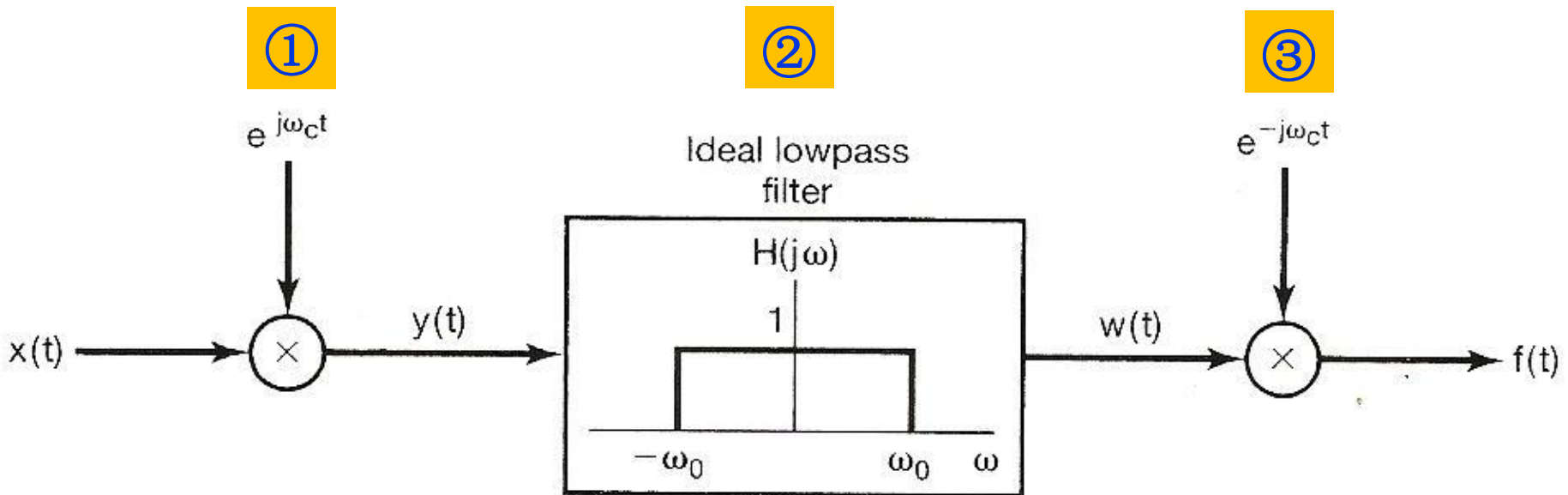
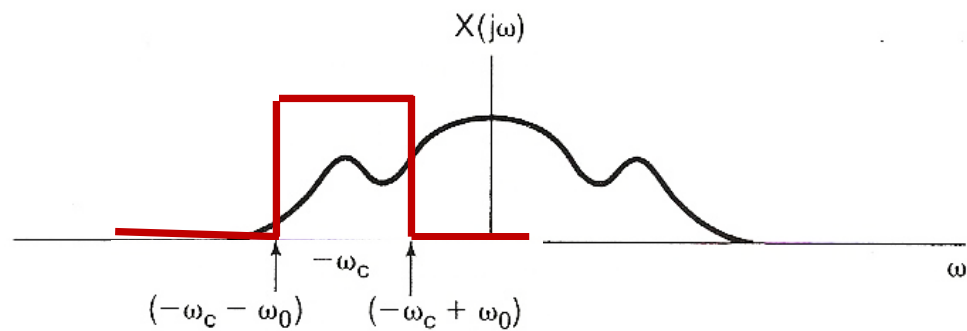
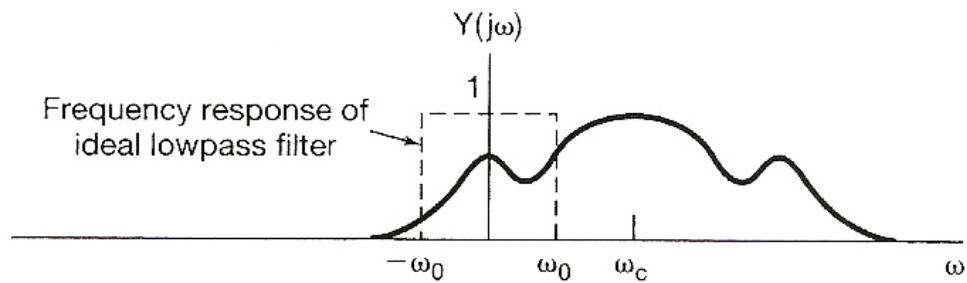


Figure 4.26 Implementation of a bandpass filter using amplitude modulation with a complex exponential carrier.

①



②



③

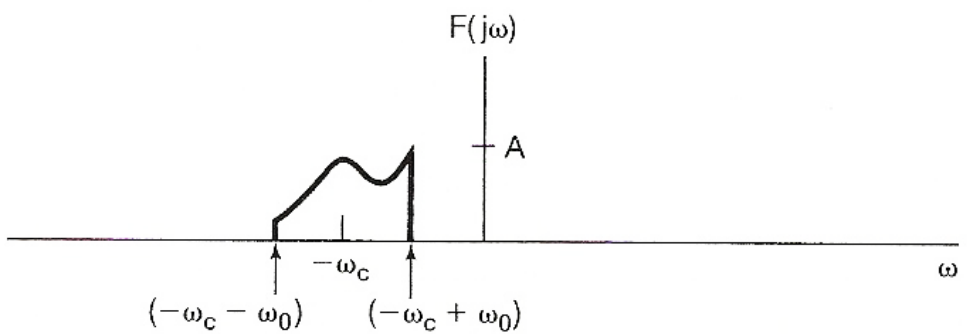
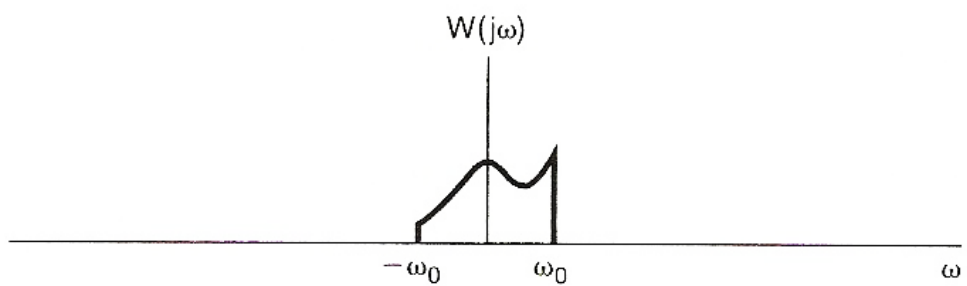


Figure 4.27 Spectra of the signals in the system of Figure 4.26.

Example: Amplitude Modulation

Example 4.21

$$r(t) = s(t) \cdot p(t) \longleftrightarrow R(j\omega) = \frac{1}{2\pi} [S(j\omega) * P(j\omega)]$$

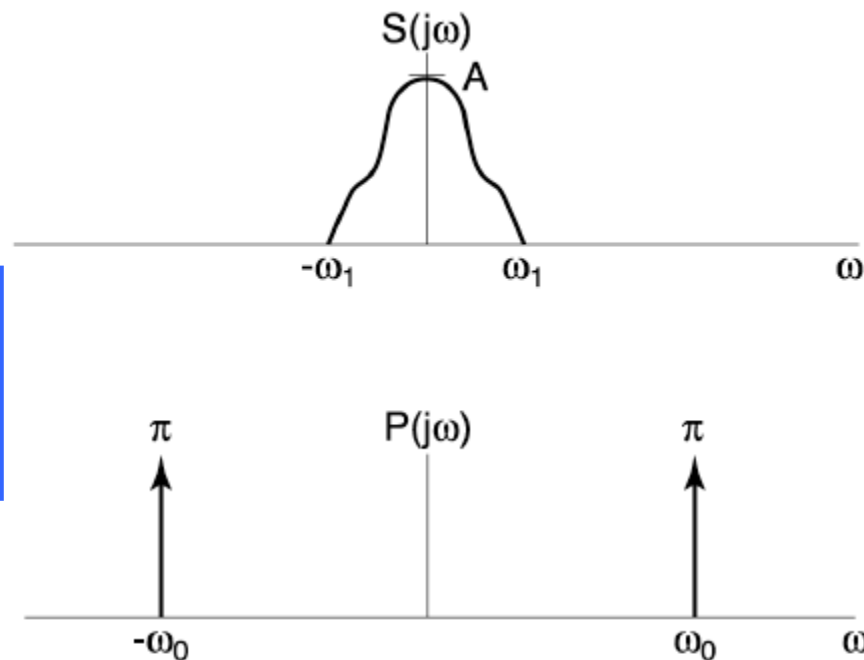
$$\text{For } p(t) = \cos \omega_0 t \longleftrightarrow P(j\omega) = \pi \delta(\omega - \omega_0) + \pi \delta(\omega + \omega_0)$$

$$R(j\omega) = \frac{1}{2} S(j(\omega - \omega_0)) + \frac{1}{2} S(j(\omega + \omega_0))$$

(cont.)

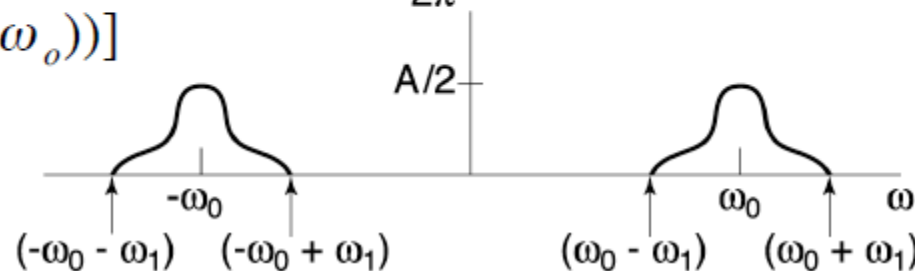
 ω_1 : bandwidth

$r(t) = s(t) \cdot \cos(\omega_o t)$
 Amplitude
 modulation (*AM*)



$$R(j\omega) = \frac{1}{2} [S(j(\omega - \omega_o)) + S(j(\omega + \omega_o))]$$

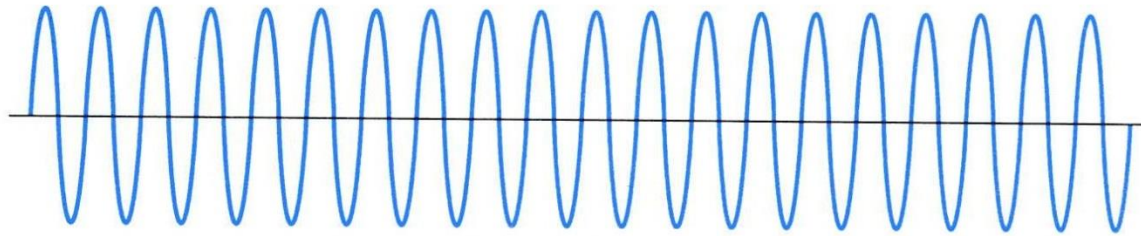
$$R(j\omega) = \frac{1}{2\pi} [S(j\omega) * P(j\omega)]$$



Drawn assume
 $\omega_o - \omega_1 > 0$
i.e. $\omega_o > \omega_1$

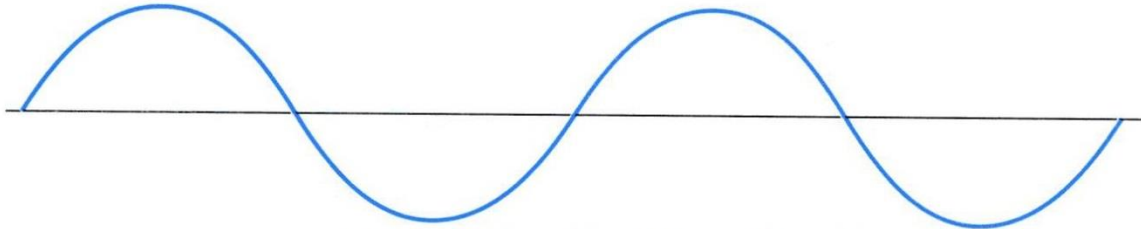
(cont.)

$p(t)$



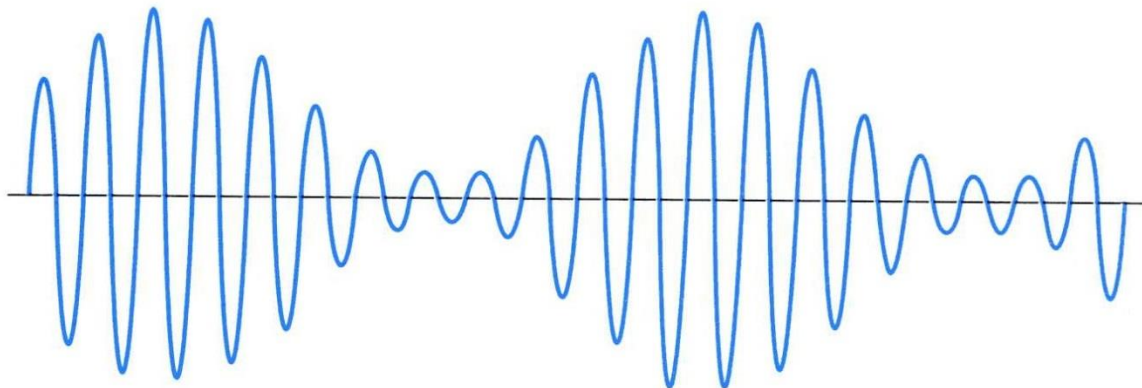
Carrier Signal

$s(t)$



Modulating Sine Wave Signal

$r(t)$



Amplitude Modulated Signal

ironbark.xtelco.com.au

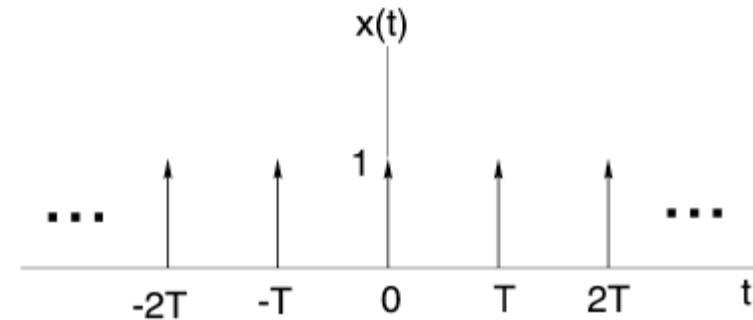
Example 4.8

$$x(t) = \sum_{n=-\infty}^{+\infty} \delta(t - nT) \quad \text{— sampling function}$$

$$x(t) \xleftrightarrow{\text{FS}} a_k = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jk\omega_o t} dt = \frac{1}{T}$$

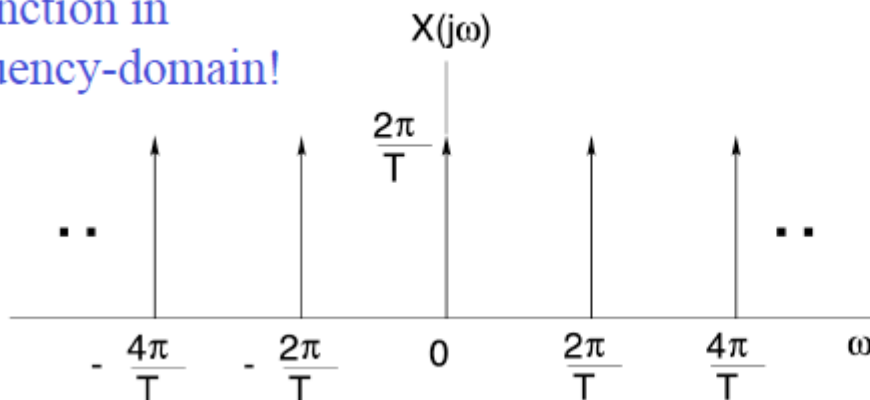
$$\Downarrow x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_o t} = \sum_{k=-\infty}^{+\infty} \frac{1}{T} e^{jk\omega_o t}$$

$$X(j\omega) = \sum_{k=-\infty}^{+\infty} \frac{2\pi}{T} \delta\left(\omega - \underbrace{\frac{k2\pi}{T}}_{k\omega_o}\right)$$



$\omega_s = 2\pi / T$: sampling frequency

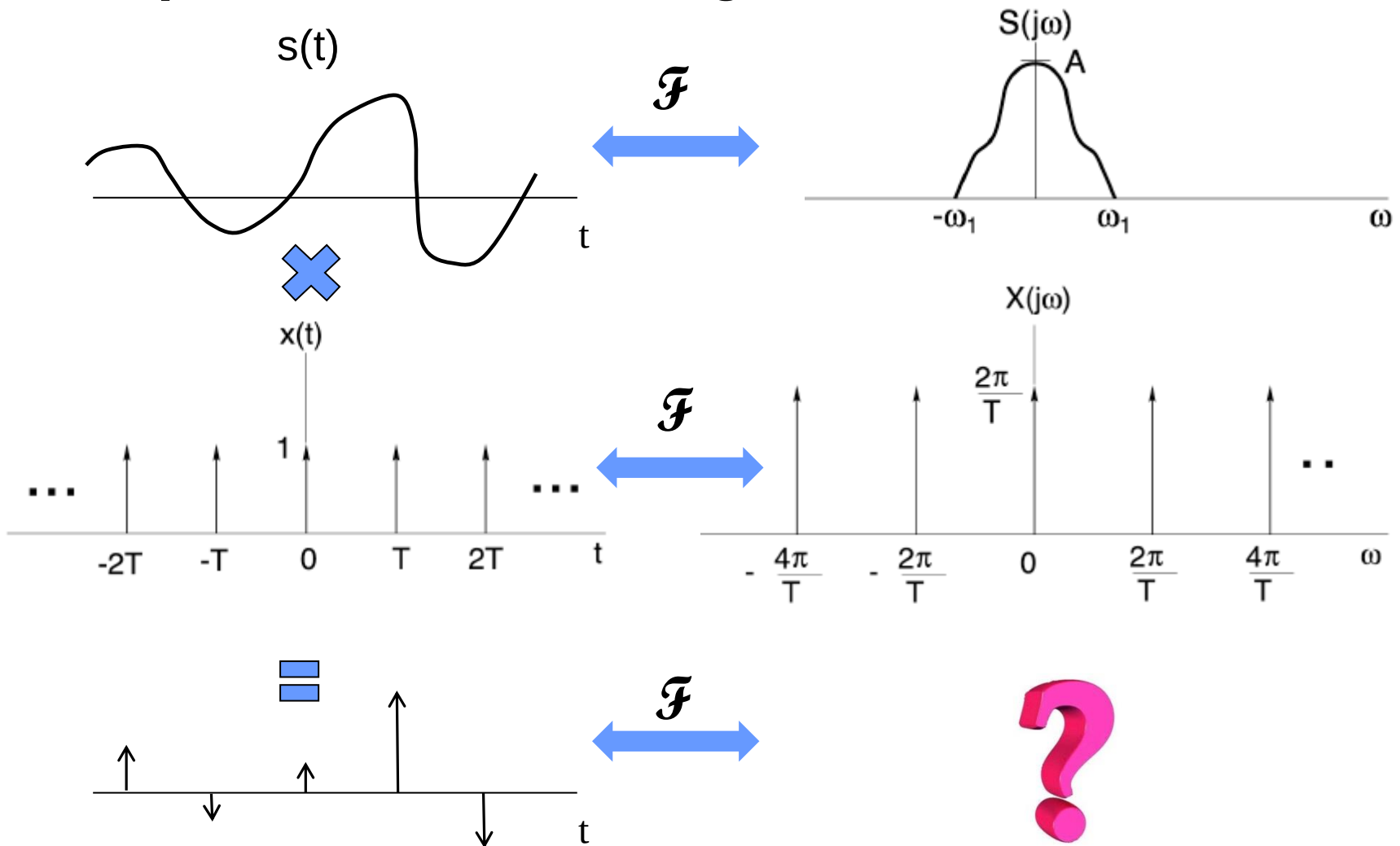
Same function in
the frequency-domain!



Note in this case, periodic
in both time domain (with
a period T) and frequency
domain (with a period
 $2\pi/T$)

Example: Sampling

- Sample a continuous signal



LTI Systems by LCCDE

(Linear-constant-coefficient differential equations)

$$\sum_{k=0}^N a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^M b_k \frac{d^k x(t)}{dt^k}$$

Using the Differentiation Property

$$\frac{d^k x(t)}{dt^k} \longleftrightarrow (j\omega)^k X(j\omega)$$

$\Downarrow$ Transform both sides of the equation

$$\sum_{k=0}^N a_k (j\omega)^k Y(j\omega) = \sum_{k=0}^M b_k (j\omega)^k X(j\omega)$$

$\Downarrow$

$$Y(j\omega) = \underbrace{\left[\frac{\sum_{k=0}^M b_k (j\omega)^k}{\sum_{k=0}^N a_k (j\omega)^k} \right]}_{H(j\omega)} X(j\omega)$$

$$H(j\omega) = \left[\frac{\sum_{k=0}^M b_k (j\omega)^k}{\sum_{k=0}^N a_k (j\omega)^k} \right]$$

Partial Fraction Expansion

Partial Fraction Expansion (No identical poles):

$$\begin{aligned} H(j\omega) &= \frac{b_M(j\omega + z_1)(j\omega + z_2) \dots (j\omega + z_M)}{a_N(j\omega + p_1)(j\omega + p_2) \dots (j\omega + p_N)} \\ &= \frac{A_1}{j\omega + p_1} + \frac{A_2}{j\omega + p_2} + \dots + \frac{A_N}{j\omega + p_N} \end{aligned}$$

Partial Fraction Expansion (with identical poles):

$$\begin{aligned} H(j\omega) &= \frac{b_M(j\omega + z_1)(j\omega + z_2) \dots (j\omega + z_M)}{a_N(j\omega + p_1)^{k_1}(j\omega + p_2)^{k_2} \dots (j\omega + p_n)^{k_n}} = \\ &= \frac{A_{1,1}}{(j\omega + p_1)^{k_1}} + \frac{A_{1,2}}{(j\omega + p_1)^{k_1-1}} + \dots + \frac{A_{1,k_1}}{(j\omega + p_1)} \\ &\quad + \dots + \\ &\quad \frac{A_{n,1}}{(j\omega + p_n)^{k_n}} + \frac{A_{n,2}}{(j\omega + p_n)^{k_n-1}} + \dots + \frac{A_{n,k_n}}{(j\omega + p_n)} \end{aligned}$$

- A stable LTI system S has the frequency response

$$H(j\omega) = \frac{j\omega + 4}{6 - \omega^2 + 5j\omega}$$

- ◆ Determine a differential equation relating the input $x(t)$ and output $y(t)$ of S
- ◆ Determine the impulse response $h(t)$ of S
- ◆ What is the output of S when the input is $x(t) = e^{-4t}u(t) - te^{-4t}u(t)$

① $\frac{Y(j\omega)}{X(j\omega)} = H(j\omega) = \frac{j\omega + 4}{6 - \omega^2 + 5j\omega}$

$Y(j\omega) \cdot (6 - \omega^2 + 5j\omega) = X(j\omega) \cdot (j\omega + 4)$

$\Updownarrow$

$\frac{d^2 y(t)}{dt^2} + 5 \frac{dy(t)}{dt} + 6y(t) = \frac{dx(t)}{dt} + 4x(t)$

② $H(j\omega) = \frac{j\omega + 4}{(2 + j\omega)(3 + j\omega)} = \frac{A}{2 + j\omega} - \frac{B}{3 + j\omega} = \frac{2}{2 + j\omega} - \frac{1}{3 + j\omega}$

$\therefore e^{-at}u(t) \longleftrightarrow \frac{1}{a + j\omega} \quad \therefore h(t) = 2e^{-2t}u(t) - e^{-3t}u(t)$

③ $\therefore te^{-at}u(t) \longleftrightarrow \frac{1}{(a + j\omega)^2}$

$\therefore X(j\omega) = \frac{1}{4 + j\omega} - \frac{1}{(4 + j\omega)^2}$

$\therefore Y(j\omega) = H(j\omega) \cdot X(j\omega) = \frac{1}{(4 + j\omega)(2 + j\omega)} = \frac{A}{4 + j\omega} - \frac{B}{2 + j\omega}$

TABLE 4.1 PROPERTIES OF THE FOURIER TRANSFORM

Section	Property	Aperiodic signal	Fourier transform
		$x(t)$ $y(t)$	$X(j\omega)$ $Y(j\omega)$
4.3.1	Linearity	$ax(t) + by(t)$	$aX(j\omega) + bY(j\omega)$
4.3.2	Time Shifting	$x(t - t_0)$	$e^{-j\omega t_0} X(j\omega)$
4.3.6	Frequency Shifting	$e^{j\omega_0 t} x(t)$	$X(j(\omega - \omega_0))$
4.3.3	Conjugation	$x^*(t)$	$X^*(-j\omega)$
4.3.5	Time Reversal	$x(-t)$	$X(-j\omega)$
4.3.5	Time and Frequency Scaling	$x(at)$	$\frac{1}{ a } X\left(\frac{j\omega}{a}\right)$
4.4	Convolution	$x(t) * y(t)$	$X(j\omega)Y(j\omega)$
4.5	Multiplication	$x(t)y(t)$	$\frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\theta)Y(j(\omega - \theta))d\theta$
4.3.4	Differentiation in Time	$\frac{d}{dt} x(t)$	$j\omega X(j\omega)$
4.3.4	Integration	$\int_{-\infty}^t x(t)dt$	$\frac{1}{j\omega} X(j\omega) + \pi X(0)\delta(\omega)$
4.3.6	Differentiation in Frequency	$tx(t)$	$j \frac{d}{d\omega} X(j\omega)$
4.3.3	Conjugate Symmetry for Real Signals	$x(t)$ real	$\begin{cases} X(j\omega) = X^*(-j\omega) \\ \Re\{X(j\omega)\} = \Re\{X(-j\omega)\} \\ \Im\{X(j\omega)\} = -\Im\{X(-j\omega)\} \\ X(j\omega) = X(-j\omega) \\ \angle X(j\omega) = -\angle X(-j\omega) \end{cases}$
4.3.3	Symmetry for Real and Even Signals	$x(t)$ real and even	$X(j\omega)$ real and even
4.3.3	Symmetry for Real and Odd Signals	$x(t)$ real and odd	$X(j\omega)$ purely imaginary and odd
4.3.3	Even-Odd Decomposition for Real Signals	$x_e(t) = \mathcal{E}\{x(t)\}$ [x(t) real] $x_o(t) = \mathcal{O}\{x(t)\}$ [x(t) real]	$\Re\{X(j\omega)\}$ $j\Im\{X(j\omega)\}$
4.3.7	Parseval's Relation for Aperiodic Signals		
	$\int_{-\infty}^{+\infty} x(t) ^2 dt = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) ^2 d\omega$		

Table 4.2

Basic Fourier Transform Pairs

Signal	Fourier transform	Fourier series coefficients (if periodic)
$\sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$	$2\pi \sum_{k=-\infty}^{+\infty} a_k \delta(\omega - k\omega_0)$	a_k
$e^{j\omega_0 t}$	$2\pi \delta(\omega - \omega_0)$	$a_1 = 1$ $a_k = 0$, otherwise
$\cos \omega_0 t$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$	$a_1 = a_{-1} = \frac{1}{2}$ $a_k = 0$, otherwise
$\sin \omega_0 t$	$\frac{\pi}{j}[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$	$a_1 = -a_{-1} = \frac{1}{2j}$ $a_k = 0$, otherwise
$x(t) = 1$	$2\pi \delta(\omega)$	$a_0 = 1$, $a_k = 0$, $k \neq 0$ (this is the Fourier series representation for any choice of $T > 0$)
Periodic square wave		
$x(t) = \begin{cases} 1, & t < T_1 \\ 0, & T_1 < t \leq \frac{T}{2} \end{cases}$ and $x(t + T) = x(t)$	$\sum_{k=-\infty}^{+\infty} \frac{2 \sin k\omega_0 T_1}{k} \delta(\omega - k\omega_0)$	$\frac{\omega_0 T_1}{\pi} \operatorname{sinc}\left(\frac{k\omega_0 T_1}{\pi}\right) = \frac{\sin k\omega_0 T_1}{k\pi}$
$\sum_{n=-\infty}^{+\infty} \delta(t - nT)$	$\frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta\left(\omega - \frac{2\pi k}{T}\right)$	$a_k = \frac{1}{T}$ for all k
$x(t) \begin{cases} 1, & t < T_1 \\ 0, & t > T_1 \end{cases}$	$\frac{2 \sin \omega T_1}{\omega}$	—
$\frac{\sin Wt}{\pi t}$	$X(j\omega) = \begin{cases} 1, & \omega < W \\ 0, & \omega > W \end{cases}$	—
$\delta(t)$	1	—
$u(t)$	$\frac{1}{j\omega} + \pi \delta(\omega)$	—
$\delta(t - t_0)$	$e^{-j\omega t_0}$	—
$e^{-at} u(t), \operatorname{Re}\{a\} > 0$	$\frac{1}{a + j\omega}$	—
$te^{-at} u(t), \operatorname{Re}\{a\} > 0$	$\frac{1}{(a + j\omega)^2}$	—
$\frac{t^{n-1}}{(n-1)!} e^{-at} u(t), \operatorname{Re}\{a\} > 0$	$\frac{1}{(a + j\omega)^n}$	—

$$e^{-at} u(t) \longleftrightarrow \frac{1}{a + j\omega}$$

$$te^{-at} u(t) \longleftrightarrow \frac{1}{(a + j\omega)^2}$$